

Introducing Concept of Indirect Proportion

Steps:

1. Analyse the 2 scenarios below:

- If a painter takes 1 day to paint a house, how long will 2 painters take? (More or less time?)



- If a car travels at 60 km/h, it takes 3 hours to reach a destination, how long will it take if the speed is 90 km/h? (More or less time?)



Discuss: "What happens when there is more workers/speed?"

2. What is meant by indirect proportion?

Definition: Two quantities are said to be in indirect proportion or inversely proportional if when one quantity increases, the other decreases at the same rate.

Demonstrating Concept of Indirect Proportion

3. Class Activity

- Colour the drawing provided individually and in groups of 3.
Discuss the time taken by 1 student vs time taken by 3 students to finish colouring the same drawing.
- A student walks slowly vs a student walks quickly across the classroom.
Discuss the time taken by each.

Profession-Related Problem Solving.

Example 1

A painter takes 4 days to paint a house.

- (a) How long will 2 painters take?
(b) How long will 4 painters take?

Number of painters			
Time taken (days)	4	2	1
	1 painter $\rightarrow$ 4 days	1 painter $\rightarrow$ 4 days 2 painters $\rightarrow 4 \div 2$ $= 2$ days	1 painter $\rightarrow$ 4 days 4 painters $\rightarrow 4 \div 4 = 1$ day

Example 2

2 masons take 6 hours to build a wall. How long will 3 masons take?

Number of masons			
Time taken (hours)	12	6	4
	2 masons $\rightarrow$ 6 hours 1 mason $\rightarrow 6 \times 2 = 12$ hrs	2 masons $\rightarrow$ 6 hours	2 masons $\rightarrow$ 6 hours 1 mason $\rightarrow 6 \times 2 = 12$ hrs 3 masons $\rightarrow 12 \div 3 = 4$ hrs

Example 3

3 gardeners take 4 hours to clean the garden. How long will 2 gardeners take?

Number of gardeners			
Time taken (hours)	12	6	4
	3 gardeners $\rightarrow$ 4 hours 1 gardener $\rightarrow 4 \times 3$ $= 12$ hrs	3 gardeners $\rightarrow$ 4 hours 1 gardener $\rightarrow 4 \times 3$ $= 12$ hrs 2 gardeners $\rightarrow 12 \div 2$ $= 6$ hrs	3 gardeners $\rightarrow$ 4 hours

G. Assessment

Divide students in pairs or groups of 3 to solve the following profession-related problems. Hints are given to struggling students.

Question 1

If 1 dressmaker sews a dress in 8 hours, how long will 2 dressmakers take?

Number of dressmakers		
Time taken (hours)	8	_____
	1 dressmaker → 8 hours	2 dressmakers →

Question 2

A baker prepares all the different cakes for an event in 12 hours; how long will 5 bakers need to prepare all the cakes for the same event? (Answer in both hours and minutes)

1 baker → 12 hours

5 bakers →



Question 3

2 cooks prepare a meal in 3 hours; how long will 3 cooks take to prepare the same meal?

Number of cooks			
Time taken (hours)	_____	3	_____
	2 cooks → 3 hours 1 cook →	2 cooks → 3 hours	2 cooks → 3 hours 1 cook → 3 cooks →

Question 4

4 men take 6 hours to repair a road. How long will 8 men take to do the same work?

4 men → 6 hours

1 man →

8 men →



Question 5

If 6 carpenters build a table in 2 days, how long will 3 carpenters take?

6 carpenters → 2 days

1 carpenter →

3 carpenters →



H. Extension of activity